BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, AJK, MIRPUR

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021 • Paper Code: 21/01

Retyped faithfully from the original board paper — same questions, no changes.

MATHEMATICS (PART –II) (OBJECTIVE PART)

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 21/01

Note: Note:- Write your Roll No. in space provided. Over writing, cutting, using of lead pencil will result in loss of marks. All questions are to be attempted.

Q1. (Marks (20)) Each question has four possible answers, Tick (√) the correct answer.

Q1. $d/dx\ (\sin \sqrt{x}) =$

(A) $\cos \sqrt{x}$

(B) $-\cos \sqrt{x}$

(C) $(-1/(2\sqrt{x}))\cos \sqrt{x}$

(D) $(1/(2\sqrt{x}))\cos \sqrt{x}$

Q2. $d/dx\ (\sinh^{-1}x) =$

(A) $1/\sqrt{(1+x^2)}$

(B) $-1/\sqrt{(1+x^2)}$

(C) $1/\sqrt{(1-x^2)}$

(D) $-1/\sqrt{(1-x^2)}$

Q3. $\int(\sin^2x + \cos^2x)dx =$

(A) $\sin x + \cos x + c$

(B) $\cos 2x + \sin 2x + c$

(C) $x + c$

(D) $(1/2)x^2 + c$

Q4. $\int(\tan^{-1}x \cdot 1/(1+x^2))dx =$

(A) $(1/2)(\tan^{-1}x)^2 + c$

(B) $\ln(\tan^{-1}x) + c$

(C) $2(\tan^{-1}x)^2 + c$

(D) $(\tan^{-1}x)^2 + c$

Q5. $\int(1/(x\sqrt{(x^2-a^2)}))dx =$

(A) $(1/a)\tan^{-1}(x/a) + c$

(B) $(1/a)\text{Sec}^{-1}(x/a) + c$

(C) $\tan^{-1}(x/a) + c$

(D) $\text{Sec}^{-1}(x/a) + c$

Q6. $\int_0^1 1/(1+x^2)\,dx =$

(A) $\pi/2$ (B) $\pi/6$

(C) $\pi/3$ (D) $\pi/4$

Q7. The function $f(x) = 3x/(x^2+1)$ is _____ function.

(A) Even (B) Odd

(C) Linear (D) Quadratic

Q8. If $g(x) = 1/x^2$, $x \neq 0$ then $gog(x) =$

(A) 1 (B) x^2

(C) x^4 (D) $1/x^4$

Q9. The derivative of $(x^3 + 2x^2)/x^3$ equals

(A) $2/x^2$ (B) $-2/x^2$

(C) $1/(2x^2)$ (D) $-1/(2x^2)$

Q10. $d/dx\,(x - 1/x) =$

(A) $1 - 1/x$ (B) $1 + 1/x$

(C) $1 + 1/x^2$ (D) $1 - 1/x^2$

> Note: (P.T.O)

Q11. The circle $x^2 + y^2 + 2gx + 2fy + c = 0$, passes through the origin if;

(A) $c = 0$ (B) $c = -1$

(C) $c = 1$ (D) $c = 2$

Q12. Focus of the parabola $x^2 = -16y$ is;

(A) $(0,4)$ (B) $(0,-4)$

(C) $(4,0)$ (D) $(-4,0)$

Q13. Foci of the hyperbola $x^2/a^2 - y^2/b^2 = 1$, are

(A) $(\pm a,0)$ (B) $(0,\pm a)$

(C) $(\pm ae,0)$ (D) $(0,\pm ae)$

Q14. The unit vector in the direction of $v = 2i - j$ is

(A) $(2i - j)/3$ (B) $(2i - j)/\sqrt{3}$

(C) $(2i - j)/5$ (D) $(2i - j)/\sqrt{5}$

Q15. If the vectors $2i + 4j - 7k$ and $2i + 6j + xk$ are perpendicular, then $x =$

(A) 8 (B) 4

(C) 2 (D) 1

Q16. $2i \times 2j.k =$

(A) 2 (B) 4

(C) 8 (D) 1

Q17. The distance of the point $(-1,3)$ from x-axis is;

(A) 1 (B) -1

(C) 3 (D) -3

Q18. Slope of the straight line $ax + by + c = 0$, is;

(A) a/b (B) $-a/b$

(C) b/a (D) $-b/a$

Q19. The lines l_1 and l_2 with slopes m_1 and m_2 respectively are perpendicular if;

(A) $m_1 = m_2$ (B) $m_1m_2 = 1$

(C) $m_1m_2 = -1$ (D) $m_1 + m_2 = 0$

Q20. The point (1,2) lies in the solution of the inequality;

(A) $2x + y > 5$ (B) $2x + y < 3$

(C) $2x + y > 6$ (D) $x + 3y > 5$

> Note: (The End)

MATHEMATICS PAPER : PART – II (SUBJECTIVE PART)

Time Allowed: 2:30 Hours | **Maximum Marks:** 80 | **Paper Code:** 21/01

> Note: NOTE:- Attempt any TWENTY FIVE (25) short questions in all selecting eight from Q. 2 and Q. 3 each and nine from Q. 4. (25 x 2 = 50)

> Note: SECTION – I

Q2. (Marks (2 x 8 = 16)) Write short answers of any eight parts.

(i) Find the domain and range of $f(x) = \sqrt{(x^2 - 9)}$

(ii) Determine whether $f(x) = \text{Sin } x + \text{Cos } x$ is even or odd.

(iii) For the functions $f(x) = 1/\sqrt{(x-1)}$ and $g(x) = 1/x^2$, $x \neq 0$ find fog(x) and gof(x)

(iv) Evaluate Lim $(n\to\infty)$ (1 + 3/n)^(2n)

(v) Differentiate $y = 1/\sqrt{x}$ by definitions.

(vi) Differentiate $(x^2 - 1)/(x^2 - 3)$ w.r.t x

(vii) $y = (3x^2 - 2x + 7)^6$ find dy/dx

(viii) tan y(1 + tan x) = (1 − tan x) show that dy/dx = −1

(ix) Differentiate $(\ln x)^x$ w.r.t x

(x) Find y_2 if y = (2x + 5)^(3/2)

(xi) Find the extreme values of the function $f(x) = 5x^2 - 6x + 2$

(xii) Find dy/dx if y = e^(−2x) sin 2x

Q3. (Marks (2 x 8 = 16)) Write short answers of any eight parts.

(i) Find dy if $y = x^2 - 1$ and x changes from 3 to 3.02

(ii) Evaluate the integral $\int(1-\sqrt{x})^2/\sqrt{x}\,dx$, (x > 0)

(iii) Evaluate $\int 1/(x \ln x)\,dx$

(iv) Evaluate the integral $\int e^x(1/x + \ln x)\,dx$

(v) Evaluate $\int(3x+1)/(x^2-x-6)\,dx$

(vi) Evaluate $\int_1^2 (x^2+1)\,dx$

(vii) Find the area between the x-axis and the curve $y = x^2 + 1$ from x = 1 to x = 2

(viii) Solve the differential equation dy/dx = −y

(ix) Show that the point A(0,2), B(√3,−1) and C(0,−2) are vertices of a right triangle.

(x) Two points P and O′ are given in xy-coordinate system. Find the XY-coordinates of P refered to the translated axis. P(−6,−8); O′(−4,−6)

(xi) Find equation of line through (−5,−3) and (9,−1)

(xii) Determine the value of P such that the lines, 2x−3y−1=0, 3x−y−5=0 and 3x+py+8=0 meet at a point.

> Note: P.T.O

> Note: (2)

Q4. (Marks (2 x 9 = 18)) Write short answers of any nine parts.

(i) Graph the solution set of given linear inequality in xy-plane 5x − 4y ≤ 20

(ii) Find an equation of the circle with ends of a diameter at (−3,2) and (5,−6)

(iii) Write an equation of the parabola with given elements. Focus (−3,1); directrix x − 2y − 3 = 0

(iv) Find an equation of directrix of given parabola; $x^2 = 4(y - 1)$

(v) Find eccentricity of $y^2/16 - x^2/9 = 1$

(vi) Find Foci and vertices of hyperbola. $9x^2 - 12x - y^2 - 2y + 2 = 0$

(vii) Find a unit vector in the direction of the vector $v = 2i + 6j$

(viii) Find a vector whose magnitude is 2 and is parallel to $-i + j + k$

(ix) Find the direction cosines for the vector: $v = 3i - j + 2k$

(x) Find the cosine of the angle θ between u and v: $u = [2,-3,1]$, $v = [2,4,1]$

(xi) Compute $a \times b$, check your answer by showing that a is perpendicular to $a \times b$. $a = -4i + j - 2k$, $b = 2i + j + k$

(xii) Find volume of the parallelepiped determined by; $u = i + 2j - k$; $v = i - 2j + 3k$, $w = i - 7j - 4k$

(xiii) A force $F = 4i - 3k$, passes through the point $A(2,-2,5)$. Find the moment of F about the point $B(1,-3,1)$

> Note: SECTION – II

> Note: Note:- Attempt any three questions from this section. All questions carry equal Marks. (10x3=30)

Q5. (Marks (10)) (a) Evaluate Lim $(\theta \to 0)$ $(\tan\theta - \sin\theta)/\sin^3\theta$ (05)

Q. (Marks (05)) (b) If $x = a\cos^3\theta$, $y = b\sin^3\theta$ then show that $a\,dy/dx + b\tan\theta = 0$

Q6. (Marks (10)) (a) Show that $\int\sqrt{(a^2-x^2)}\,dx = (a^2/2)\sin^{-1}(x/a) + (x/2)\sqrt{(a^2-x^2)} + c$ (05)

Q. (Marks (05)) (b) Find h such that the points A(h,1), B(2,7) and C(−6,−7) are vertices of a right triangle with right angle at the vertex A.

Q7. (Marks (10)) (a) Evaluate $\int_0\hat{}(\pi/4)\ 1/(1+\sin x)\,dx$ (05)

Q. (Marks (05)) (b) Graph the feasible region of system of linear inequalities. $2x-3y \le 6$, $2x+3y \le 12$, $x \ge 0$, $y \ge 0$. Also find corner points.

Q8. (Marks (10)) (a) Find the volume of Tetrahedron with vertices. (0,1,2)(3,2,1)(1,2,1) and (5,5,6) (05)

Q. (Marks (05)) (b) Write an equation of circle passing through the points A(5,6), B(−3,2), C(3,−4)

Q9. (Marks (10)) (a) Find the point on the curve $y = x^2 - 1$ that is closest to the point (3,−1) (05)

Q. (Marks (05)) (b) Write an equation of the parabola with given elements. Axis parallel to y-axis, the points (0,3),(3,4) and (4,11) lie on the graph.

> Note: (The End)